Andrew J. Stier

The Santa Fe Institute
Omidyar Postdoctoral Fellow
Emergent Political Economies Postdoctoral Fellow

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Appointments:

The Santa Fe Institute. October 2023 – October 2026
Complexity Postdoctoral Fellow
Omidyar and EPE Postdoctoral Fellow

Education:

University of Chicago
Psychology, Integrative Neuroscience Ph.D., October 2018-August 2023
M.A. March 2022
Ph.D. August 2023

University of Chicago
B.A. Physics & Mathematics, specialization Astrophysics. June 2016

Awards/Honors:

Lightning Talk Winner: Balance Between Community Fragmentation and Nodal Functional Stability in Brain Networks Transitioning Between Cognitive States. *Networks 2021 Conference.*

Nathan Sugarman Award for Excellence in Undergraduate Research. June 2016

Funding:

Lead PI. Specific or General: Empirical Measurements of Category Formation in Urban Inhabitants. *The Santa Fe Institute Lou Schuyler Seed Grant Fund Program.* **$14,987**. *1/26-9/26*

Lead PI. Collaborative Research: Understanding attitude formation at different spatial scales. NSF-2438783. *Social Psychology Program*. **$398,883**. Other personnel: Marc Berman, James Holehouse, Mahzarin Banaji. *7/25-6/28*

Published Research:

*** Corresponding Author**

***Stier, A. J.**, & Ogbunugafor, C. B. (2025). Associations between racial bias and political identity in American cities often diverge from common narratives. *npj Complexity*, *2*(1), 20.

Jeong, H. J., Reimann, G. E., Durham, E. L., Archer, C., **Stier, A. J.**, Moore, T. M., ... & Kaczkurkin, A. N. (2024). Early life stress and functional network topology in children. *Developmental Cognitive Neuroscience*, *66*, 101367.

***Stier, A. J**., Sajjadi, S., Karimi, F., Bettencourt, L. M., & Berman, M. G. (2024). Implicit racial biases are lower in more populous more diverse and less segregated US cities. *Nature Communications*, *15*(1), 961.

Mummaneni, A., Kardan, O., **Stier, A. J.**, Chamberlain, T. A., Chao, A. F., Berman, M. G., & Rosenberg, M. D. (2023). Functional brain connectivity predicts sleep duration in youth and adults. *Human Brain Mapping*, *44*(18), 6293-6307.

***Stier, A. J.,** Cardenas-Iniguez, C., Kardan, O., Moore, T. M., Meyer, F. A., Rosenberg, M. D., ... & Berman, M. G. (2023). A pattern of cognitive resource disruptions in childhood psychopathology. *Network Neuroscience*, *7*(3), 1153-1180.

Schertz, K. E., Kotabe, H. P., Meidenbauer, K. L., Layden, E. A., Zhen, J., Bowman, J. E., ... & Berman, M. G. (2023). Nature's path to thinking about others and the surrounding environment. *Journal of Environmental Psychology*, *89*, 102046.

Durham, E. L., Ghanem, K., **Stier, A. J.**, Cardenas-Iniguez, C., Reimann, G. E., Jeong, H. J., ... & Kaczkurkin, A. N. (2023). Multivariate analytical approaches for investigating brain-behavior relationships. *Frontiers in Neuroscience*, *17*, 1175690.

Jeong, H. J., Reimann, G., Durham, E. L., **Stier, A. J.**, Moore, T., Dupont, R. M., ... & Kaczkurkin, A. (2023). 138. Functional Network Topology and Environmental Stressor. *Biological Psychiatry*, *93*(9), S150.

***Stier, A. J.** (2023). Urban Psychology: A Framework for Ecologically Relevant Investigations of Human Behavior.

Kardan, O., **Stier, A. J.**, Cardenas-Iniguez, C., Schertz, K. E., Pruin, J. C., Deng, Y., ... & Rosenberg, M. D. (2022). Differences in the functional brain architecture of sustained attention and working memory in youth and adults. *Plos Biology*, *20*(12), e3001938.

Durham, E. L., **Stier, A. J.**, Cardenas-Iniguez, C., Reimann, G., Jeong, H. J., Dupont, R. M., ... Others. (2022). P440. Investigation of Multivariate Associations Between Brain Volume and Dimensions of Psychopathology in Children. *Biological Psychiatry*, *91*(9), S265–S266.

Reimann, G., **Stier, A. J.**, Moore, T. M., Durham, E. L., Jeong, H. J., Cardenas-Iniguez, C., ... Others. (2022). P68. Atypical Network Properties During Rest and Task Performance in Youth With ADHD Symptoms: A Bifactor Model Approach. *Biological Psychiatry*, *91*(9), S114–S115.

***Stier, A. J.**, Schertz, K. E., Rim, N. W., Cardenas-Iniguez, C., Lahey, B. B., Bettencourt, L. M. A., & Berman, M. G. (2022). Reply to Huth et al.: Cities are defined by their spatially aggregated socioeconomic networks. *Proceedings of the National Academy of Sciences*, *119*(2), e2119313118.

Zhuang, C., Meidenbauer, K. L., Kardan, O., **Stier, A. J.**, Choe, K. W., Cardenas-Iniguez, C., ... Berman, M. G. (2022). Scale Invariance in fNIRS as a Measurement of Cognitive Load. *Cortex*.

Meidenbauer, K. L., Niu, T., Choe, K. W., **Stier, A. J.**, & Berman, M. G. (2021). Mouse

movements reflect personality traits and task attentiveness in online experiments. *Journal of Personality*.

Brandtner, C., Bettencourt, L. M. A., Berman, M. G., & **Stier, A. J.** (2021). Creatures of the state? Metropolitan counties compensated for state inaction in initial US response to COVID-19 pandemic. *PloS one*, *16*(2), e0246249.

***Stier, A. J.**, Schertz, K. E., Rim, N. W., Cardenas-Iniguez, C., Lahey, B. B., Bettencourt, L. M. A., & Berman, M. G. (2021). Evidence and theory for lower rates of depression in larger US urban areas. *Proceedings of the National Academy of Sciences*, *118*(31), e2022472118.

***Stier, A. J.**, Berman, M. G., & Bettencourt, L. (2021). Early pandemic COVID-19 case growth rates increase with city size. *npj Urban Sustainability*, *1*(1), 1–6.

Moore, T. M., Kaczkurkin, A. N., Durham, E. L., Jeong, H. J., McDowell, M. G., Dupont, R. M., … Others. (2020). Criterion validity and relationships between alternative hierarchical dimensional models of general and specific psychopathology. *Journal of abnormal psychology*, *129*(7), 677.
Berman, M. G., **Stier, A. J.**, & Akcelik, G. N. (2019). Environmental neuroscience. *American Psychologist*, *74*(9), 1039.

Submitted Manuscripts:

Hartle, H., Wolpert, D., **Stier, A. J.**, Kempes, C. P., & Manzano, G. (2024). Work extraction with feedback control using limited resources. *arXiv preprint arXiv:2407.05507*. (**In Press, PRR**)

***Stier, A.J.** (2025). Implicit Biases Change More Slowly in Larger Cities. https://doi.org/10.31234/osf.io/gp64e_v1 (**In Revision, PNAS**)

***Stier, A. J.**, Sajjadi, S., Bettencourt, L., Karimi, F., & Berman, M. G. (2022). Effects of Racial Segregation on Economic Productivity in US Cities. *arXiv preprint arXiv:2212.03147*. (**In Revision**, **Nature Cities**)

*^Lee, F. M., Berman, M. G., **Stier, A. J.**, & Bainbridge, W. A. (2024). Navigating Memorability Landscapes: Hyperbolic Geometry Reveals Hierarchical Structures in Object Concept Memory. *BioRxiv*, 2024-09. (^co-senior-author, *corresponding author, submitted, PNAS Nexus)

Preprints:

***Stier, A.J.**, Shi, N., Al Kontar, R., Giusti, C., Berman, M. G. (2025)**.** ALBATROSS: Cheap Filtration Based Geometry via Stochastic Sub-Sampling.

***Stier, A. J.**, & Berman, M. (2024). Evidence for Distance Dependence and Out-Group Differences in Emotion Recognition. https://doi.org/10.31234/osf.io/mb2sr

Science Communication:

Stier, A. J. (2021, December 21). *Why life is faster but depression is lower in bigger cities: Psyche ideas*. Psyche. https://psyche.co/ideas/why-life-is-faster-but-depression-is-lower-in-

bigger-cities

Invited Talks:

The Influence of Cities and Culture on Human Behavior. *Harvard Psychology Brown Bag.* November 2025

The Influence of Cities and Culture on Human Behavior. *Oxford INET Seminar.* November 2025

Using Multiplayer Video Games to Understand Computation in Human Society. *Santa Fe Institute Working Group: Energy, Information, and Computation in Human Societies.* December 2023

Collaborative Work Extraction With Limited Resources. *Santa Fe Institute Working Group: Stochastic Thermodynamics and Computer Science Theory. June* 2024

A Contingent Search for Non-Contingent Patterns of Human Behavior. *Santa Fe Institute Working Group: Contingency in Complex Systems. July* 2024

Urban and Biological Scaling. *Santa Fe Institute Complexity Global School. July* 2024

Urban Psychology. *Santa Fe Institute GAINs School. October* 2024

Meetings Organized:

Identifying General Socioeconomic and Environmental Factors Effecting Human Decision-Making and Behavior. *SFI Working Group. October* 2025 *(Co-organized with Luis Bettencourt)*. https://www.santafe.edu/events/identifying-general-socioeconomic-and-environmental-factors-affecting-human-decision-making-and-behavior

Studying the Micro-Foundations of Urban Interactions with Large Scale Data. *SFI Micro-Working Group*. May 2025 (*Co-organized with Christa Brelsford)* https://www.santafe.edu/events/studying-the-micro-foundations-of-urban-interactions-with-large-scale-data

Towards the Integration of Socio-Cultural and Individual Difference Approaches to Implicit Attitudes. *SFI Working Group.* May 2025 *(Co-organized with Mahzarin Banaji)* *https://www.santafe.edu/events/towards-the-integration-of-socio-cultural-and-individual-difference-approaches-to-implicit-attitudes*

Building Scaling Models of Nervous Systems II. *SFI Micro-Working Group. January* 2025 *(Co-organized with Chris Kempes and Ignacio Arroyo)* *https://www.santafe.edu/events/building-scaling-models-of-nervous-systems-ii*

Exploring Humanity and Society Through Video Games. *SFI Micro-Working Group with independent video game developers.* February 2025 *(Co-organized with Cris Moore)* *https://www.santafe.edu/events/exploring-humanity-and-society-through-video-games*

Building Scaling Models of Nervous Systems. *SFI Micro-Working Group. January* 2025 *(Co-organized with Chris Kempes and Ignacio Arroyo)* *https://www.santafe.edu/events/building-scaling-models-of-nervous-systems*

Mentorship:

Masters Students:

Fiona Lee, The University of Chicago, M.A. (graduated June 2023)
Rudolf Hu, The University of Chicago, M.A. (graduated June 2023)

Undergraduates:

Federica Sanchez y Carillo Bernal, Undergraduate at National Autonomous University of Mexico (UNAM)
The Santa Fe Institude Summer REU Program (2024)
Maya Ordonez, The University of Chicago (2021-2023)

Training and Skills:
Biological Sciences Division Quantitative Biology Boot Camp. Marine Biological Laboratory. September 2018

Brain Connectivity Workshop, May 2021

ABCD Longitudinal Modeling Workshop, June/July 2021

Urban Science Certificate, Mansueto Institute for Urban Innovation. Awarded, June 2022

Brain Connectivity Workshop, June 2022

SFI Complexity GAINs Workshop: *Disintegration of societies: Quantitative modeling of complex socio-behavioral systems*, July 2022

Algebraic Topology/Persistent Homology, fMRI (pre-processing, multivariate analysis, MVPA, functional connectivity), Bayesian modeling (Hierarchical Bayes and Empirical Bayes), Java, Javascript, Qualtrics, Prolific, R, Scala, Matlab, Fortran, Labview, Mathematica, C++, Python, Haskell and Root programming languages. MongoDB, Spark, HDFS, Kafka

Teaching:
One-on-one course of machine learning for mental health diagnosis and treatment prediction, Horizon Labs, June-September 2020

Sensation & Perception, T.A., Winter 2021
Cognitive Psychology, T.A., Spring 2021
Experimental Design Course Prep, T.A., 2021
Experimental Design 1, T.A., Winter 2021
Experimental Design 2, T.A., Spring 2022

SFI Complexity Global School, Faculty, July 2024
SFI GAINs School: Ecology, Faculty, October 2024
SFI Complexity Global School, Faculty, July 2025

Other Professional Experience:
Goldman Sachs
Credit Risk & Alternative Investments and Manager Selection, Technology Analyst. July 2016-May 2018

University of Chicago

CTA Gamma Ray Telescope Camera Development Lab (PI Scott Wakely). June 2014-June 2016
HELIX project (PI Scott Wakely). January 2016 – June 2016

Service:
University of Chicago Men's Crew Captain; *2014-2016*
Parent-Alumni Liaison; *2013-2014*
SFI Omidyar Fellow Selection Committee; *2023-Present*
Slice of Science talk series organizer; *2023-Present*
NSF Social Psychology Review Panel; *November, 2025*

Journal Reviews
Nature Cities
Nature Computational Sciences
Network Neuroscience
PLOS Biology
Science Advances
PLOS One
npj Complexity
npj Urban Sustainability
American Journal of Psychology

Select Media Coverage:

https://www.apa.org/research-practice/conduct-research/hidden-association

https://www.bloomberg.com/news/articles/2021-08-02/how-big-cities-help-fight-depression

https://www.thetimes.com/uk/healthcare/article/living-in-a-big-city-can-help-fight-depression-cmrrqkqfx